

Low Voltage and High Speed 1Xnm 1T1C FE-RAM with Ultra-Thin 5nm HZO

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Abstract— World-first 1Xnm half-pitch FE-RAM with 8Gb density was fabricated, and operation was confirmed. The conventional FE-RAM maximizes 2Pr by adjusting the capacitor plate voltage according to the data. In this study, despite using a fixed capacitor plate voltage, we showed that memory operation is possible even at a low voltage of $\pm 0.6V$ by using the pinched hysteresis of 5nm-thick ultra-thin HZO. We measured the switching speed by changing the write time from 5ns to 80ns. 70% of the total polarization can be switched within 20ns tWR (like the DRAM), and the remaining 30% responds to a wide range of write time between 20 and 80ns. For improving the switching speed, it is necessary to reduce bulk defects or design schemes such as Vcore overdrive.

I. INTRODUCTION

Since the ferroelectricity had been found in HfO₂, ferroelectric HfO₂ has attracted many attentions as a strong candidate for the emerging memory due to the process easiness and compatibility. In particular, HfZrO (HZO) is being studied as a material candidate for metal-ferroelectric-metal (MFM) capacitors because it has a low thermal budget and high step coverage that can be uniformly deposited on the high aspect ratio capacitor of DRAM. Conventional studies using the MFM capacitors have been mainly conducted on HZO with a thickness of 10 nm or more on a 2D flat capacitor. Manufacturing high-density FE-RAM requires characteristics study in higher aspect ratio 3D capacitor structures with thinner HZO for scalability. The manufacture of high-performance FE-RAM requires research on polarization switching speed. Therefore, we studied polarization characteristics and speed characteristics of ultra-thin HZO by FE-RAM fabricated using 1Xnm DRAM cell array.

II. THE DESIGN OF EXPERIMENT

A. Process integration

8Gb 1T1C FE-RAM is fabricated using 1Xnm DRAM technology. In DRAM MIM Cap, the insulator was changed to HZO, and thickness splits were carried out from 5nm to 7.5nm to verify the characteristics of HZO according to thickness. For FE-RAM operation, the DRAM Sense amplify was changed to Sense Amplifier With Adaptive Reference (SAWAR). Figure 1 shows the key process steps and cross section HR-TEM images of the fabricated device. The anneal process for crystallization of HZO after plate TiN deposition is carried out for a relatively long time with soak anneal. Unlike other studies conducted in

2D flat capacitors, high aspect ratio 3D structure such as DRAM capacitor make it difficult for RTA to apply sufficient thermal evenly throughout the entire capacitor. The results of crystal analysis with HR-TEM images through FFT pattern analysis show that the HZO layer consists of orthorhombic phase poly crystal with various crystal directions and very little monoclinic phase.

B. Test methodology for HZO characteristics investigation

The PUND measurement through the test pattern composed of 10k HZO MFM capacitor array and function test through fully functional FE-RAM chip was conducted. We measured the hysteresis curve of HZO capacitors through the PUND measurement. We evaluated speed property through function test because the test pattern has too high a capacitance to measure the speed characteristics. Figure 2 shows the PUND trapezoidal waveform, the current and the P-V hysteresis curve integrated with the current. From this P-V curve, the 2Pr, which is the remnant polarization, was extracted. Characteristics according to voltage and HZO thickness can be inferred through changes in 2Pr and current peak.

Figure 3 shows the schematic of SAWAR, the timing diagram of the Read/Write operation of FE-RAM, and the hysteresis state. The Read sequence is as follows. First, when BL is pre-charged to Vss (=0V) to use switching charge and Cell Tr. is turned on, BL rises to VH by switching charge + dielectric charge in case of 1 data and BL increases to VL by only dielectric charge in case of 0 data. Second, by setting /BL to Vref between VH and VL, during SA operation, BL rises to Vcore (1 data voltage) or falls to Vss (0 data). Third, the capacitor will be re-written. Finally, after tWR(write recovery time), BL and /BL are pre-charged to Vblp(=Vcp=capacitor plate voltage), and Cell Tr. will be turned off. VH-VL (ΔV) can be measured by scanning the Vref for every 1 and 0 data, respectively and specifying the point where the failure occurs. 2Pr can be calculated from the following formula: $2Pr/(C_{BL}+C_{DE}) = \Delta V$, when CBL is total BL capacitance and CDE is non-switching capacitance of FE-CAP. We can use ΔV instead of 2Pr because CBL, CDE does not change.

III. THE RESULTS AND DISCUSSION

A. Operation voltage and HZO thickness

Figure 4 shows the P-V hysteresis curve and current measurement according to the applied voltage of the 5nm and 7nm thick HZO MFM capacitors, respectively. From measuring the 5 nm HZO MFM capacitor at $\pm 1.5 V$, we can see

four current peaks. This result means that the curve is not a pure ferroelectric curve, but a part of anti-ferroelectric is mixed, called pinched hysteresis. On the other hand, the 7nm HZO shows more pure ferroelectric characteristics. The pinched hysteresis is because very thin HZO stabilizes the tetragonal phase as well as the orthorhombic phase, resulting in an anti-ferroelectric and ferroelectric together. [1] Figure 5 shows TCAD simulation results for pure ferroelectric (black line) and the pinched hysteresis (red line). The pinched hysteresis has a lower energy barrier than ferroelectric and has a stable point at 0V. Therefore, it has lower V_c (coercive voltage) and three current peaks when switching negative to positive, vice versa. The simulation results are consistent with the experimental results. The P-V curve is a negative shift based on 0v, not symmetrical. This asymmetry is due to the difference in the work function between the bottom electrode and the top electrode. The bottom TiN is oxidized during the HZO deposition process and has a lower work function than the top TiN[2]. Due to this asymmetry, the polarization of anti-ferroelectric is not eliminated at 0V, and FE-RAM operation is possible using this remnant polarization. Figure 4(d) shows that, unlike 5nm HZO, 7nm HZO has almost no 2Pr at 0.6V because the operating voltage cannot exceed V_c . Figure 6 shows 2Pr according to the operating voltage and HZO thickness. At low voltage, 5nm HZO offers a larger 2Pr value. As the operating voltage increases, 2Pr of the pure ferroelectric is greater than the pinched hysteresis if the operating voltage is greater than V_c and polarization switching is sufficient.

There is a way to increase the asymmetry by widening the work function difference of the top and bottom electrodes in order to boost 2Pr. It has been reported that applying different electrode materials to each bottom and top electrode makes an anti-ferroelectric loop available in ferroelectric-like behavior due to differences in work functions.[3] However, it is not easy to find the right materials that can be deposited by ALD in a 3D cap with a high aspect ratio. Another method is asymmetric operation around 0V. Figure 7 shows the concept of asymmetric operation. A larger voltage in the positive direction and a smaller voltage in the negative direction while maintaining the overall voltage would allow a larger switching region(green) to increase 2Pr.(Fig 7 (c)) Figure 8 shows the results of measurements with asymmetric operating voltage, with +0.72/-0.48 V having a polarization greater than ± 0.6 V.

By using ultra-thin HZO and asymmetry of electrode work function, it is possible to obtain the 2Pr value required for operation without driving the plate voltage at low operating voltage, which is advantageous for low power and high-speed memory manufacturing. However, low 2Pr of pinched hysteresis is a disadvantage in terms of density. We expect to overcome the low 2Pr by using asymmetric operation voltage, development of top/bottom electrode materials, and optimization of HZO composition.

B. Wafer test and speed characteristic

Figure 9(a) shows ΔV according to V_{core} and tWR. The ΔV increases when V_{core} increases because increasing the voltage at either end of the capacitor increases the FE-CAP's

2Pr. The PUND test results can also confirm this result examined earlier. Measuring ΔV by increasing the tWR from 5ns to 80ns shows that ΔV increases relative to the tWR and saturation around 65ns. This wide distribution means that not all domains in HZO have the same switching speed but a wide speed characteristic distribution ranging from 5 ns to 80 ns. According to previous studies, the switching speed of FE-HfO₂ is reported from several ns to several hundreds of ns, and the fastest switching speed was reported in the HZO thin film with a switching time of 1 ns at a single crystal HZO.[4] Figure 9(b) shows the shmoo plot of ΔV with tWR ranging from 5 to 80 ns and V_{cp} ranging from 0.6 to 0.9 V. We conducted these tests using the conditions of DRAM's wafer test program. We prove the hypothesis that asymmetric operation voltage increases 2Pr by applying V_{cp} asymmetrically in real chip operation. Figure 10 shows the ΔV portion for each tWR section. More than 70% of them operate under 20ns, and 30% need to be given more than 20ns of write time to be switched. In addition, it can be seen that speed characteristics improve when V_{core} increases from 1.2V to 1.8V. This is different from the usual case, in which the increasing of V_{core} requires a larger voltage swing, so the write requires a longer time. The NLS model can explain this speed characteristic.[5] Unlike the single-crystal ferroelectric has a fast switching speed, in the HZO thin film with a poly-crystalline, the growth rate of the domain wall is prolonged due to many defects. Figure 1 shows that our HZO film has a polycrystal structure with various crystal orientations. In order to improve the switching speed, it is possible to consider using an overdrive scheme that instantaneously applies a voltage higher than V_{core} during the write operation and improves defects inside the thin film.

IV. CONCLUSION

In this study, we showed that FE-RAM operation is possible even at low operating voltage using 2Pr from pinched hysteresis of the ultra-thin HZO capacitor and the difference in the work function of the top/bottom electrodes. We evaluated the speed characteristics by changing the write time in the write/read operation, and We confirmed that more than 70% of total cells operate under 20ns, and 30% of them need more than 20ns of write time to be switched. Therefore, research is required in order to improve 2Pr and speed up switching for manufacturing high density, high speed and low power FE-RAM.

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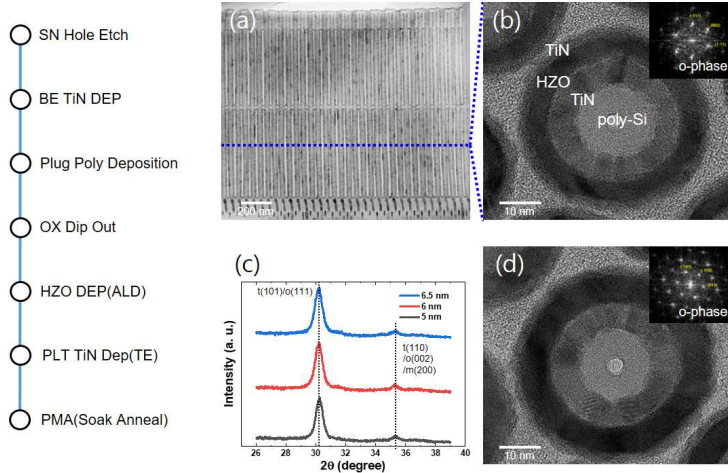


Fig. 1. Key process integration steps and (a) cross-section image of 1z nm MFM storage node, HR TEM image and FFT pattern of (b) 5 nm and (d) 7 nm thick HZO, (c) XRD intensity

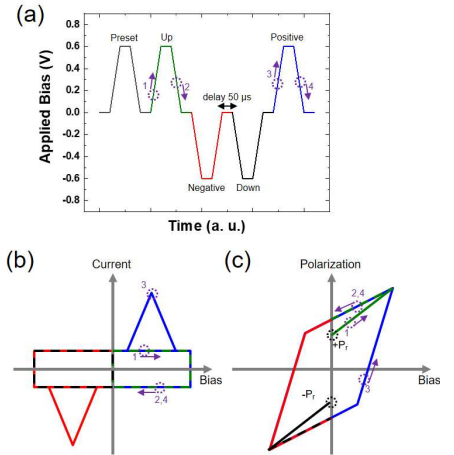


Fig. 2. Schematic of PUND measurement: (a) pulse train as a function of time, (b) transient current-V and (c) P-V hysteresis curve as results of the PUND measurement.

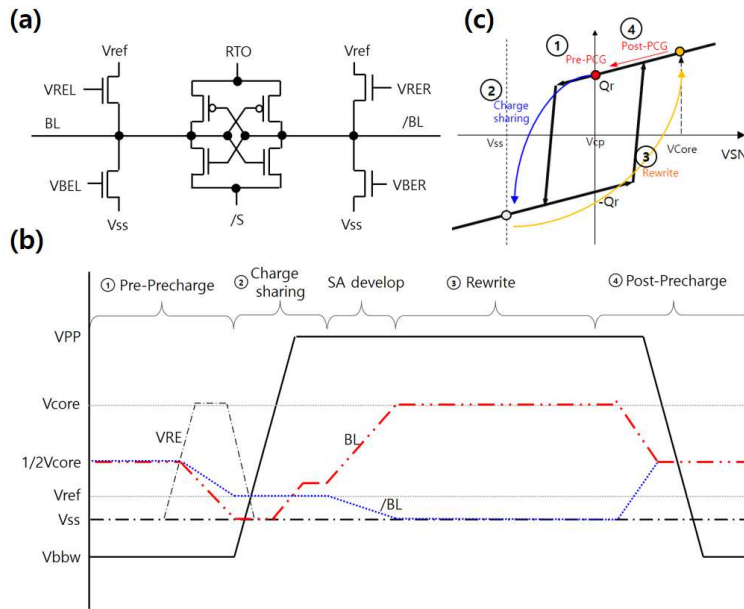


Fig. 3. (a) SAWAR (Sense Amplifier With Adaptive Reference) schematic, (b) FE-RAM 1 Read timing diagram, (c) FE-Cap states for each stage

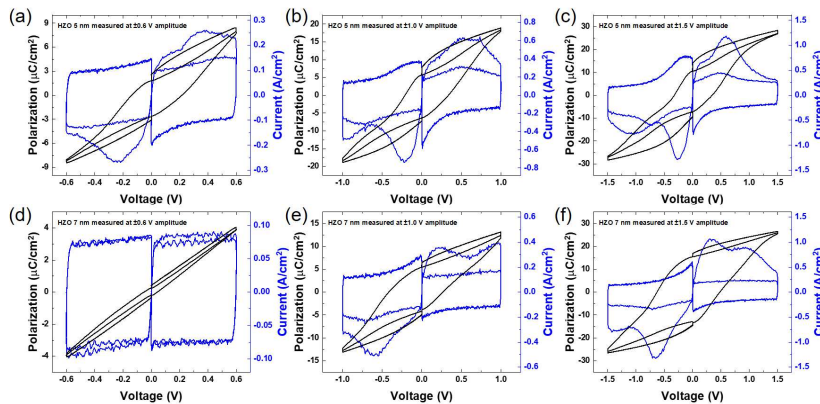


Fig. 4. The P-V and I-V hysteresis loop of 5, 7 nm-thick HZO measured at various bias conditions such as (a),(d) ± 0.6 V, (b),(e) ± 1.0 V, and (c), (f) ± 1.5 V

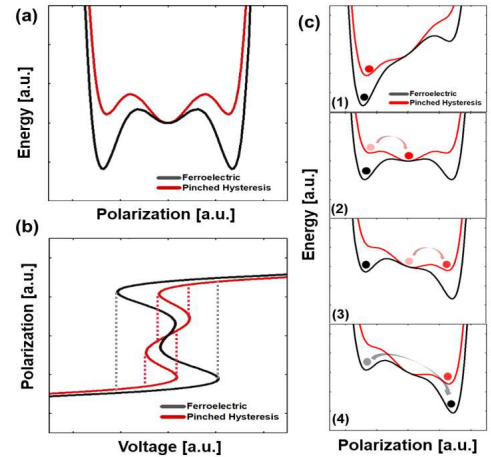


Fig. 5. characteristics of ferroelectric and pinched hysteresis material in the perspective of (a) free energy potential-polarization, (b) polarization-voltage, and (c) Free-energy potential of ferroelectric and pinched hysteresis as a function of applied field

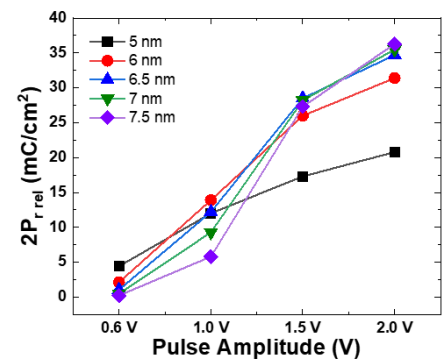


Fig. 6. The evolution of the $2P_r$ as a function of the pulse amplitude

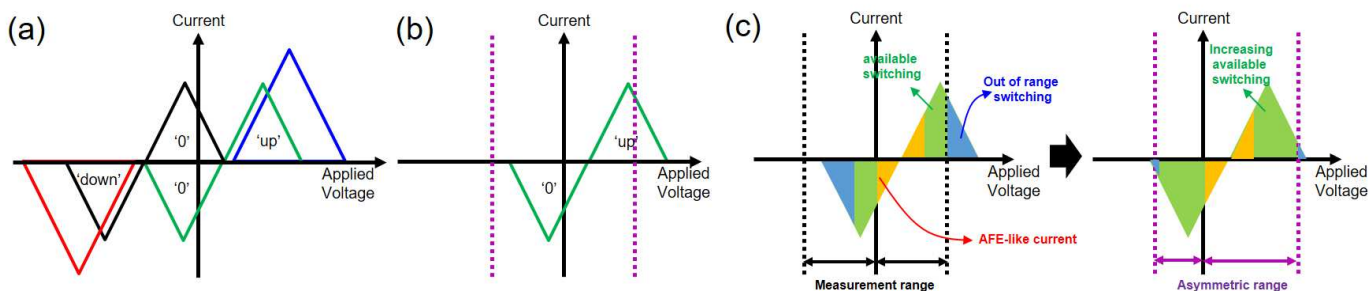


Fig. 7. Schematics of I-V characteristics of pinched hysteresis loop at different bias conditions: (a) complete switching, (b) insufficient switching, and (c) asymmetric switching.

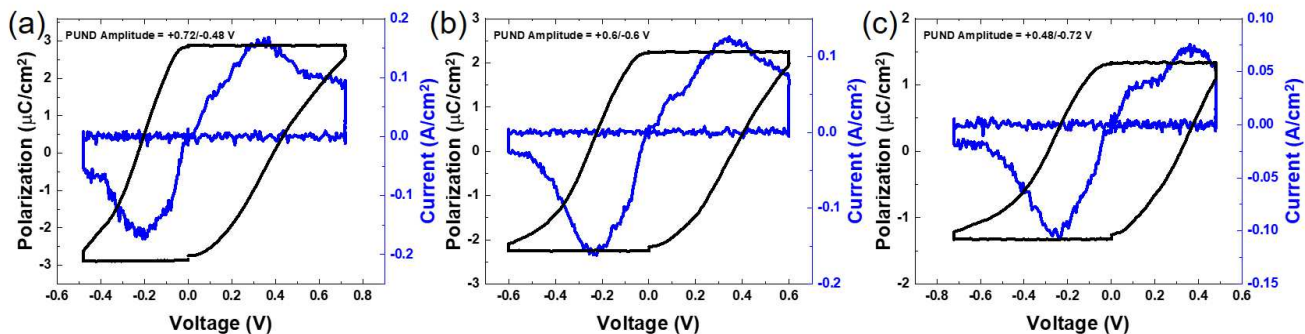


Fig. 8. The P-V and I-V hysteresis loop (switching only) of 5 nm-thick HZO measured at asymmetric bias conditions: (a) +0.72/-0.48 V, (b) ± 0.6 V, and (c) +0.48/-0.72 V.

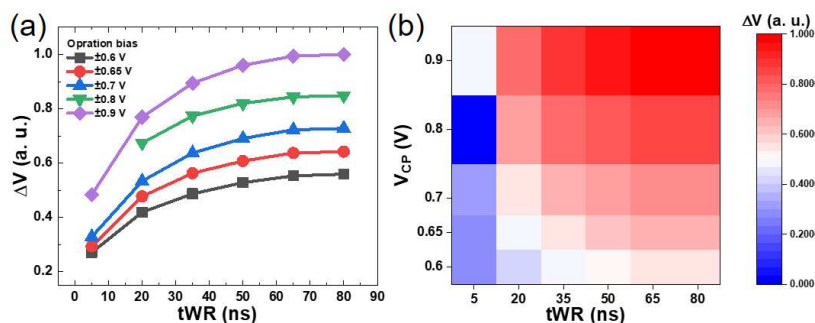


Fig. 9. (a) the evolution of the ΔV as a function of the tWR and (b) shmoo plot of ΔV tWR ranging from 5 to 80 ns and V_{cp} ranging from 0.6 to 0.9 V

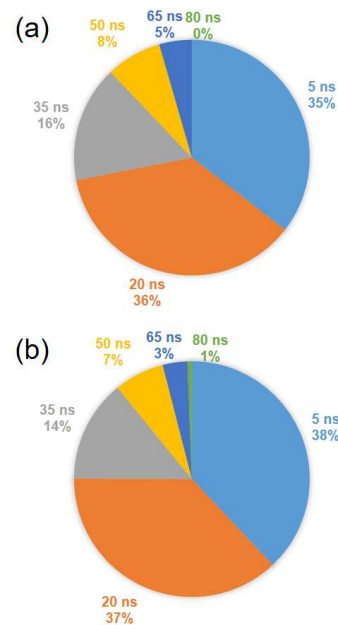


Fig. 10. Speed portion with (a) $V_{core} = 1.2$ V, (b) 1.8 V